

Topic review

Cryptococcus spp. Infection

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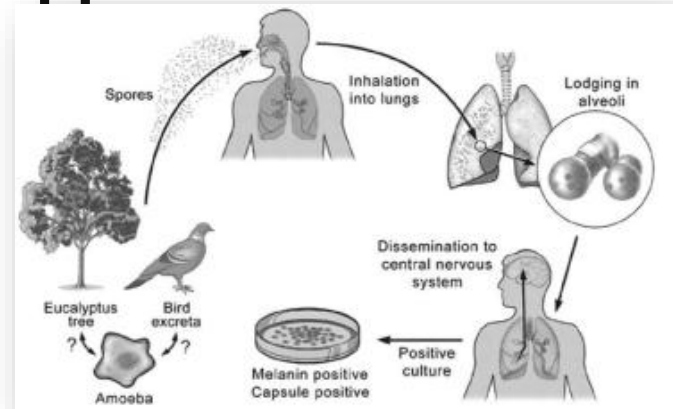


Cryptococcus spp.



Cryptococcus spp.

- Basidiomycetous fungi with more than 30 species ubiquitously distributed in the environment
- only 2 species commonly known to cause human disease
 - ***Cryptococcus neoformans***:
 - immunocompromised > immunocompetent
 - bird droppings, soil
 - ***Cryptococcus gattii***:
 - immunocompetent > immunocompromised
 - eucalyptus trees



Current classification of pathogenic *Cryptococcus* species

- Classified into **5 serotypes** (based on structural differences in the polysaccharide capsule), and **8 molecular subtypes**

Serotype	Species and Varieties	Molecular Types
A	<i>C neoformans</i> var. <i>grubii</i> ^a	VN I, VN II
B	<i>C gattii</i>	VG I, VG II, VG III, VG IV
C	<i>C gattii</i>	VG I, VG II, VG III, VG IV
D	<i>C neoformans</i> var. <i>neoformans</i>	VN IV
AD	<i>C neoformans</i>	VN III

- Approximately 95% of cryptococcal infections are caused by *C. neoformans* (serotype A) strains with the remaining 4% to 5% of infections caused by *C. neoformans* (serotype D) or *C. gattii* (serotypes B/C strains)

WHO fungal priority pathogens list (WHO FPPL)

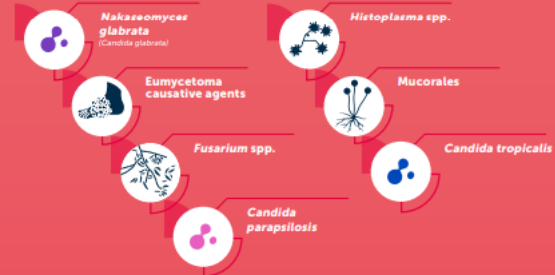
WHO fungal priority pathogens list to guide research, development and public health action, 2022

Fig. 1. WHO fungal priority pathogens list (WHO FPPL)

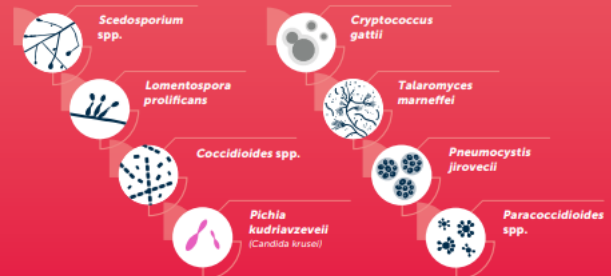
Critical Priority Group



High Priority Group

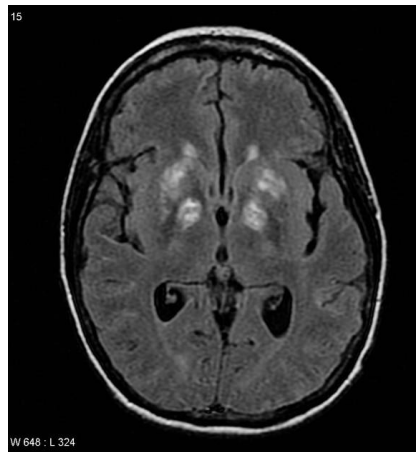


Medium Priority Group



Clinical manifestations of cryptococcosis

- Pulmonary cryptococcosis
- CNS infection (meningitis, cryptococcoma)
- Disseminated cryptococcosis
- Cutaneous cryptococcosis



Skin lesions caused by
Cryptococcus neoformans

Clinical manifestations of cryptococcosis

- *C. neoformans* species complex
 - predominant causative agent of cryptococcosis in people living with HIV
- *C. gattii* species complex
 - more commonly causes disease in people who appear immunocompetent.

- Both can cause a similarly broad range of cryptococcosis syndromes.
- *C. neoformans* has a predilection for CNS disease.
- *C. gattii* is more often associated with pulmonary disease and large cryptococcomas.

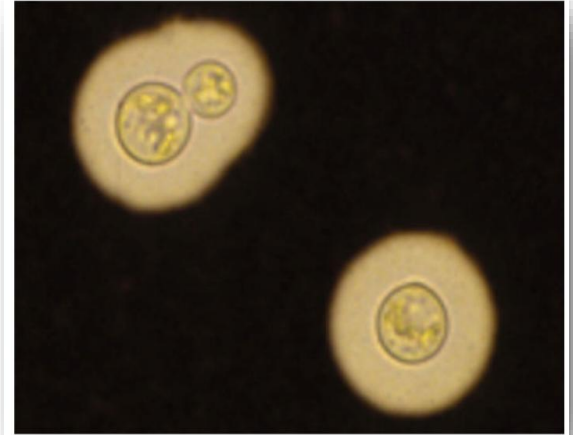
Risk factors of cryptococcal disease

- Cryptococcal disease occurs less frequently in children than in adults.
- Besides AIDS, cryptococcosis has been reported in children with a variety of underlying conditions.
- **Risk factors:**
 - Solid-organ transplant and connective tissue disease
 - Other conditions
 - Primary immunodeficiencies
 - hyper-immunoglobulin M syndrome and severe combined immunodeficiency syndrome
 - Malignancies (eg, acute lymphoblastic leukemia and sarcomas)

Laboratory diagnosis

- **Direct Examination/India Ink: CSF**

- Encapsulated yeasts
- a globular, encapsulated yeast cell with or without budding, ranging in size from 5 to 20 μm in diameter
- Sensitivity depend on fungal burden
 - 30% to 50% in non-AIDS-related cryptococcal meningitis
 - up to 80% in AIDS-related disease



CSF profile of CNS cryptococcosis in children:

- Lack in both cellularity, alteration in biochemical profiles
- Mean CSF WBC from 4-16 cells/mm³. 50% of infected children had normal CSF finding (CSF WBC < 7 cells/mm³)
- CSF protein and glucose often normal

Laboratory diagnosis

- **Culture and Identification**

- Clinical samples: **CSF, sputum, and skin biopsy, blood cultures** on routine fungal and bacterial culture media.
- Colonies are usually observed on solid agar plates after 48 to 72 hours incubation at 30°C to 35°C
 - opaque, white-to-cream colonies that may turn orange-tan or brown after prolonged incubation.
 - The mucoid appearance of the colony is related to the capsule size around the yeasts.

Cryptococcus colonies on blood agar medium



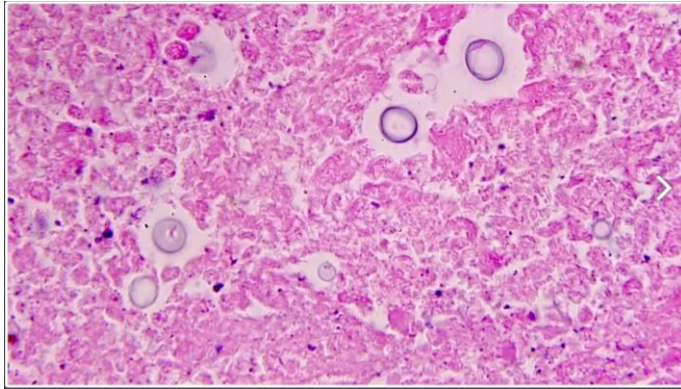
Cryptococcus colonies on Sabroud's dextrose agar medium

Laboratory diagnosis

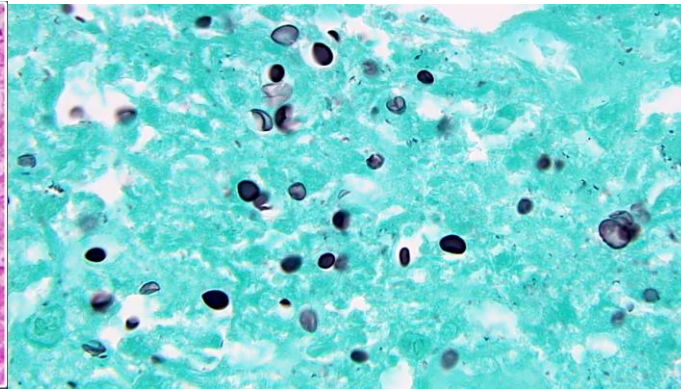
- **Cytology and Histopathology**

- Cryptococcus can be identified by histologic staining of tissues from the lung, skin, bone marrow, brain, and other organs.
- The organism is observed as a yeast that reproduces by narrow-based budding

H&E stain



GMS stain



Encapsulated variably sized yeasts with thin walls

Laboratory diagnosis

- **Serology**

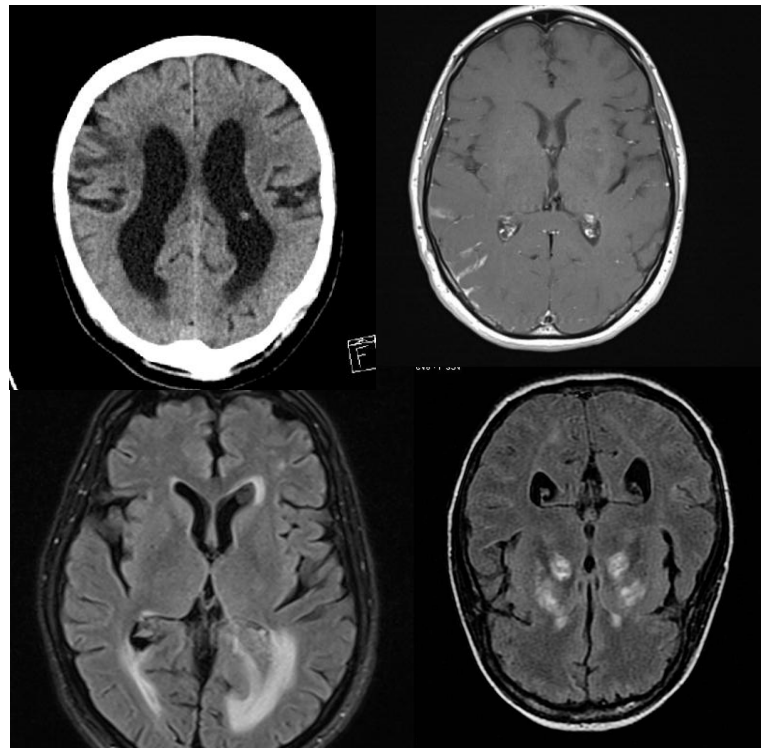
- Serologic tests for **the cryptococcal polysaccharide capsular antigen (CrAg)**, which is shed during infection.
- **Latex agglutination and enzyme immunoassay techniques**
 - Sensitivities: 93% to 100%, specificities: 93% to 98%
 - false negatives can be seen owing to;
 - a prozone effect (extremely high antigen titers), which can be overcome with dilution
 - Low fungal burden (chronic low-grade meningitis or very early stages of infection)
 - Improper specimen storage

Recommendations for yeast causing cryptococcosis and diagnostic methods

- All patients with **suspected or confirmed cryptococcosis** (including cryptococcal antigenemia) require clinical assessment for **CNS, pulmonary, and other body site involvement**.
- **Investigations for disseminated disease:**
 - **LP**
 - CSF opening pressure, glucose, protein, cell counts, microscopy, and culture and quantification of CSF cryptococcal antigen
 - **Blood for cryptococcal antigen, blood cultures, sputum**, or other affected sites
 - **Brain imaging (preferably MRI)** and chest imaging (preferably CT)

Imaging in CNS cryptococcosis

- Depend on degree of immunocompromised
- **CT brain:**
 - often **non-specific and with normal scans** seen in a significant proportion of patients (reported up to ~40%)
 - 2nd most common: diffuse atrophy (relate more to HIV dementia than cryptococcosis)
 - **Hydrocephalus**
 - Cryptococcomas (cryptococcal granuloma)
 - **Leptomeningeal and pachymeningeal enhancement**
- **MRI brain:**
 - better at assessing **dilated perivascular spaces**
 - **Basal ganglia pseudocysts**



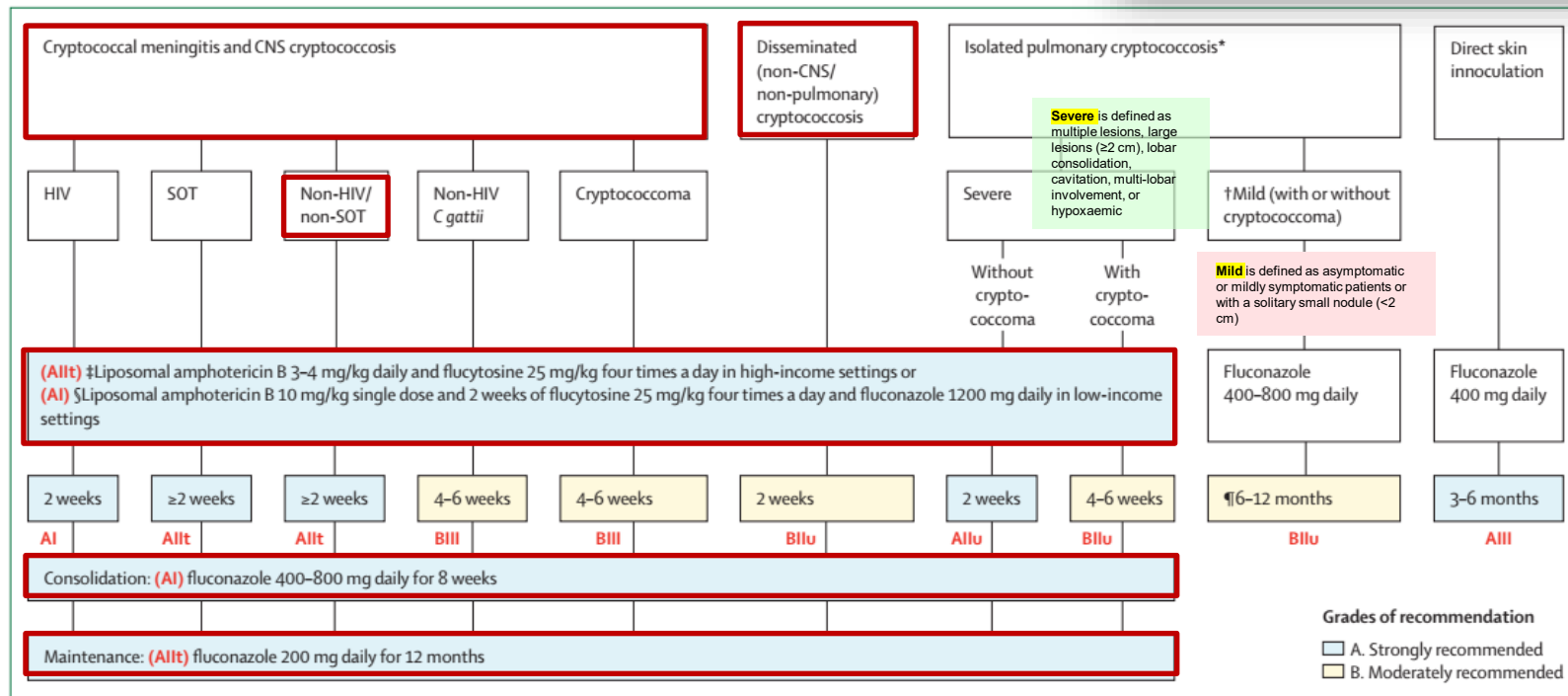
Disseminated Cryptococcosis

- **Definition of disseminated cryptococcosis:**
 - (1) positive blood culture or
 - (2) positive culture from at least two different sites.
- Disseminated cryptococcosis in children is a classic affliction associated with **HIV infection** or **primary inherited immunodeficiency disorders (PID)** with central nervous system being the most common site of dissemination.

Recommended first-line antifungal therapy by cryptococcosis syndrome (Adults)

Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM

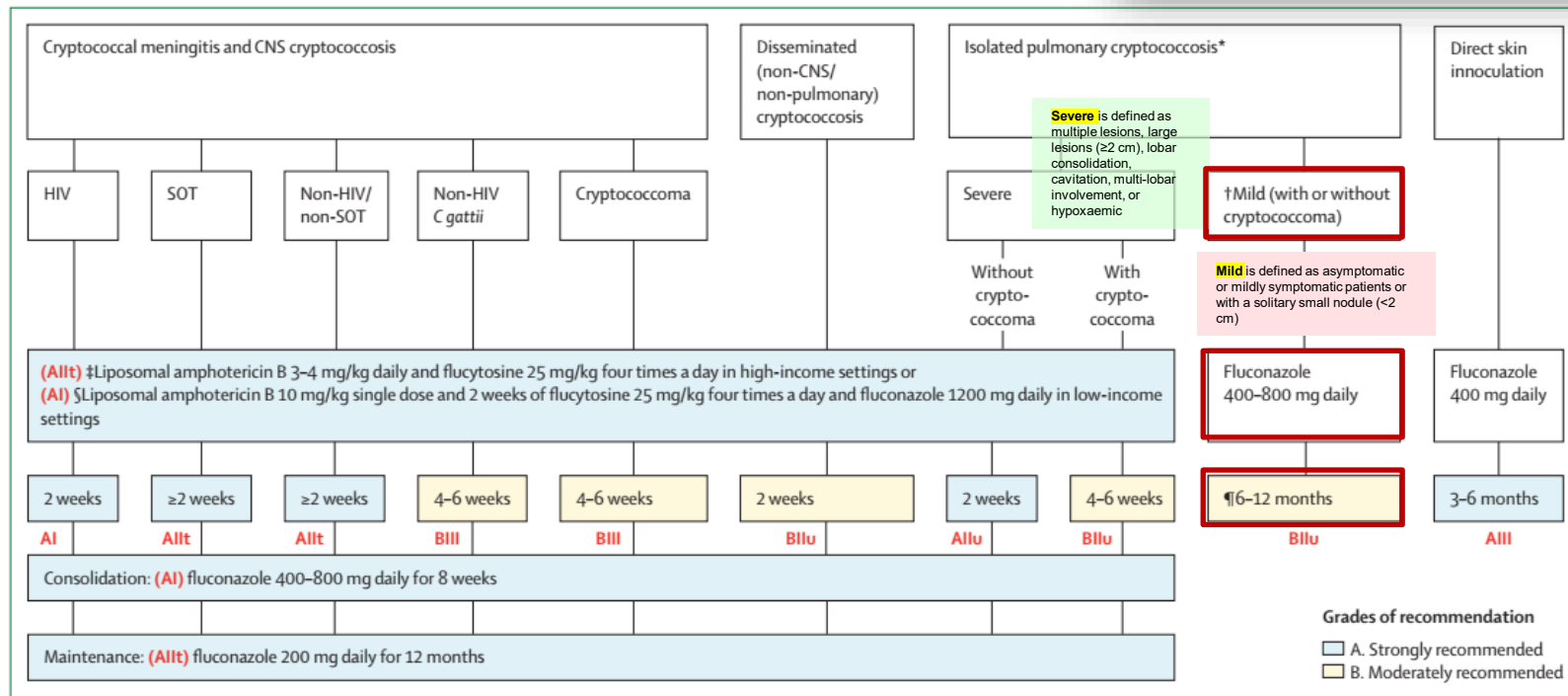
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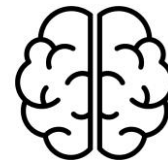
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Antifungal treatment for cryptococcal meningitis (Adult)



Patient without HIV or SOT

First-line therapies

Induction (minimum 2 weeks)

(AIII) Liposomal amphotericin B 3–4 mg/kg daily plus flucytosine 25 mg/kg four times a day

Consolidation (8 weeks)

(AIII) Fluconazole 400–800 mg daily

Maintenance (12 months)

(AIII) Fluconazole 200 mg daily

Alternative therapies

If liposomal amphotericin B is not available:

(BIII) Amphotericin B lipid complex 5 mg/kg daily plus flucytosine 25 mg/kg four times a day

If liposomal amphotericin B and amphotericin B lipid complex are not available:

(BIII) Amphotericin B 0.7–1.0 mg/kg daily plus flucytosine 25 mg/kg four times a day

If amphotericin B-based therapies are not able to be used:

(CIII) flucytosine 25 mg/kg four times a day plus fluconazole 800–1200 mg daily

(BIII) Voriconazole 200 mg twice a day (with TDM)

(BIII) Posaconazole 300 mg daily (with TDM)

(BIII) Isavuconazole 200 mg daily

(CIII) Itraconazole 200 mg twice a day (with TDM)

Grades of recommendation

- A. Strongly recommended
- B. Moderately recommended
- C. Marginally recommended

Comments:

- Recommendations in HIV patient population are also applicable
- **(AIII)** Induction therapy with liposomal amphotericin B and flucytosine should be considered for any disseminated disease or isolation from a sterile site (even in the absence of CNS manifestations)
- **(AIII)** Close monitoring of tacrolimus, cyclosporine, and sirolimus concentrations (TDM) and dose reduction of these agents are recommended when azoles are co-administered^{73,74}
- **(BIII)** Immunosuppressant doses need to be carefully adjusted to allow effective killing of yeasts but should be reduced slowly to avoid precipitating cryptococcosis-associated immune reconstitution inflammatory syndrome; consider a sequential or stepwise reduction of immunosuppressants with careful lowering of corticosteroids early and eliminating mycophenolate before considering reduction of the calcineurin inhibitors because of their direct anticytotoxic activity
- **(CIII)** In a patient treated for cryptococcosis, retransplantation or a new organ transplant can be considered, provided viable yeasts have been cleared from CSF and the patient is asymptomatic after receiving 12 months of anticytotoxic treatment

Pediatric Cryptococcal Meningoencephalitis

- Most common symptoms:
 - headache, fever, nausea, vomiting
 - Additional: altered mental status, focal neurological deficit, photophobia
- Lack of CSF inflammation in children with cryptococcal meningoencephalitis
 - CSF profiles may be at almost normal
- Clinical efficacy trials and studies to validate diagnostics tests and therapies for cryptococcosis in children are lacking
- Recommendations for diagnostic and therapeutic management have been extrapolated from studies in adult populations but dosing of antifungal agents needs particular attention for the paediatric patient.



Recommendations for **pediatric cryptococcosis**

Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM



Christina C Chang, Thomas S Harrison, Tihana A Bicanic, Matthew Chayakulkeeree, Tania C Sorrell, Adila Warris, Ferry Hagen, Andras Spenc, Rita Oladide, Nadeeh P Gowder, Sharon C Chen, Christopher H Mady, Andreas H Groll, Yee-Chun Chen, Michael S Lionakis, Alexandre Alami, Elizabeth Castañeda, Juan Llanusa, José E Vidal, Takahiro Takazono, Martin Homgöl, Jan-William Alffenaar, Jean-Pierre Gangneux, Rajesh Soman, Li-Ping Zhi, Alexandre Bonifaz, Joseph A Jarvis, Jeremy N Day, Nikolas Klimis, Jon Salmanton-Garcia, Grigory Janvries, David B Meyer, David Lawrence, Sebastian Bahr, Felix Buegenies, Brandon J McMillen, Rouven Späth, Tinsuke S Nyska, Justin Bonfield, Fabienne Carlesse, Christopher H Heath, Olusola O Ayanlow, Olga M Mashadi, Florio Quinze-Telles Filho, Mina C Housheer, Atul K Patel, Elvin Tordick, Nina Singh, Oliver A Connolly, David R Bowdare, Oliver Lortholary, Peter G Pappas, John R Perfect

Organ involvement	Treatment
CNS or disseminated diseases	Induction: (Allt) - Amb-D 1 mg/kg daily or L-Amb 3-4 mg/kg daily plus 5-flucytosine (100–150 mg/kg daily in 4 divided doses) for 2 weeks Consolidation: (Allt) - Fluconazole 12 mg/kg (max 800 mg) daily for 8 weeks. Maintenance: (Allt) for HIV-infected and immunocompromised; (BItt) for immunocompetent). - Fluconazole 6 mg/kg (max 800 mg) daily for 6–12 months.
non-CNS, but severe isolated lung disease	as for CNS disease
non-CNS, mild isolated lung disease	Fluconazole 12 mg/kg (max 800 mg) daily for 3-6 months.

Raised intracranial pressure

- Associated with a **high burden of cryptococci** → acute and chronic symptoms and signs (eg, visual and hearing loss)
- CSF outflow obstruction → improved by removal of CSF
 - **scheduled therapeutic lumbar punctures** result in substantial improvement in survival, regardless of opening pressure.
 - A **brain CT** should be done (if CNS imaging not already done) to exclude CNS outflow obstruction
- **Medical therapies:**
 - Acetazolamide, mannitol, and corticosteroids

Recommendations for raised intracranial pressure

- Acute symptomatic elevation of the intracranial pressure (≥ 20 cm of CSF)
 - Daily therapeutic lumbar punctures (removal CSF 20–30 mL), to reduce the pressure to 50% of opening pressure or to a normal pressure of ≤ 20 cm of CSF
- Perform a scheduled therapeutic lumbar puncture 48–72 h after initial lumbar puncture or 7 days, regardless of initial opening pressure
- Persistent raised symptomatic intracranial pressure
 - Surgical decompression via temporary lumbar drainage, shunting, or ventriculostomy
 - Consider VP shunt (prefer) or LP shunt (alternative) to control hydrocephalus
 - Inserts shunt after effective antifungal therapy

Cryptococcal persistence, clinical relapse, and culture positive (microbiological) relapse

- Clinical relapse can be due to a microbiological relapse, C-IRIS, raised intracranial pressure or other infective and non-infective (CNS and non-CNS) causes
- Cryptococcal antigen persists in the CSF and blood, thus it has little clinical utility in distinguishing clinical responders from non-responders.

Most cases of **culture-positive (microbiological) relapse** occur early and result from

- Inadequate or suboptimal induction therapy
- Early discontinuation of consolidation or maintenance therapy

Resistance to antifungals

- Emergence of fluconazole resistance is concerning
- Fluconazole monotherapy as induction therapy has been associated with secondary resistance
- No clinical MIC breakpoints for fluconazole against *Cryptococcus* spp
- Insufficient data to suggest that high MICs imply worse outcomes
- The epidemiological CLSI cut off values:
 - 8 ug/mL for *C. neoformans*
 - 16 ug/mL for *C. gattii*
 - 32 ug/mL for *C. deuterogattii*
- Higher than two-fold increase in MIC during treatment could suggest development of resistance

Recommendations for antifungal resistance

- For those with fluconazole resistance or emerging fluconazole resistance
 - Consider a long (eg, 4 weeks) course of induction

Induction

Amphotericin B (1 mg/kg daily) or
High dose of liposomal amphotericin B (3–6 mg/kg daily)



Flucytosine

Consolidation

- Consider amphotericin B 1 mg/kg weekly or liposomal amphotericin B 3–6 mg/kg weekly as consolidation or maintenance therapy
- Consider daily voriconazole, posaconazole, isavuconazole, or itraconazole for isolates without evidence of pan-azole resistance

Maintenance

Monitor for and minimise drug toxic effects

- Admit for the **first 1–2 weeks** is encouraged to manage the major early complications
- Use of amphotericin B and liposomal amphotericin B → pre-hydration and aggressive potassium and magnesium replacement
- Follow-up **CBC, serum creatinine and electrolyte** at least every alternate day
- **LFT** at baseline and at least weekly

Monitoring parameters: **Flucytosine**

Timing of peak	Target peak	Monitoring parameters
Obtain 2 hours post-dose within 72h after initiation or after 3 to 5 doses have been administered*	25 - 100 mg/L ¹ (40-60 mcg/mL 2 hours after the dose) ²	Serum Cr**, CBC, LFTs***

- *This lab is a send out and takes several days to result so important to time level appropriately the first time
- ** changes in renal function can have a dramatic effect on flucytosine serum concentrations given **> 90% excretion in the urine**
→ challenging when used with a nephrotoxic agent like amphotericin B
- *** **Levels > 100 mg/L increase risk of bone marrow suppression and hepatic dysfunction**

1. <https://med.stanford.edu/content/dam/sm/bugsanddrugs/documents/antimicrobial-dosing-protocols/SHC-SMUG-Flucytosine.pdf>

2. Redbook, 2024

Summary guideline for cryptococcosis

	IDSA 2010	ECMM/ ISHAM/ ASM 2024	Redbook 2024
Adult			
CNS	I: AMB 0.7-1.0 MKDay plus Flucytosine (100 MKDay) for 4 weeks C: FCZ 400-800 mg/day M: FCZ 200 mg/day	I: L-AMB 3-4 MKDay daily plus Flucytosine 25 mg/kg* 4 times - For ≥ 2 weeks (HIV, SOT, non-HIV, non-SOT) - For 4-6 weeks (non-HIV C. gatti, Cryptococcoma, severe isolated pulmonary with cryptococcoma) - For 2 weeks: Disseminated, severe isolated pulmonary without cryptococcoma C: FCZ 400-800 mg daily for 8 weeks M: FCZ 200 mg daily for 12 months	
Disseminated (non-CNS/non - pulmonary)	Same as CNS disease	Same as CNS disease	
Severe isolated pulmonary	Same as CNS disease	Same as CNS disease	
Mild isolated pulmonary	FCZ 400 mg/day for 6-12 months	FCZ 400-800 mg daily for 6-12 months	
Pediatric			
CNS	I: AmBd (1 mg/kg per day IV) plus flucytosine (100 mg/kg per day orally in 4 divided doses) for 2 weeks (for the non-HIV-infected, non-transplant C: fluconazole (10–12 mg/kg per day orally) for 8 weeks M: fluconazole (6 mg/kg per day orally)	I: AMB 1 MKDay or L-AMB 3-4 MKDay plus flucytosine 100-150 MKDay divided in 4 doses for 2 weeks C: FCZ 12 MKDay daily (max 800 mg) for 8 weeks M: FCZ 6 MKDay daily (max 800 mg) for 6-12 months	I: AMB 1 Mkday or L-AMB 5-6 Mkday + flucytosine 100 MKday for 2 weeks C: FCZ 10-12 MKDay (max 800 mg/day) for at least 8 weeks M: FCZ 6 MKDay daily (max 200 mg) at least 1 year (for HIV)
Disseminated	Same as CNS disease	Same as CNS disease	-
Severe isolated pulmonary disease	* Pneumonia: FCZ 6-12 mg/kg daily for 6-12 months	Same as CNS disease	-
Mild isolated pulmonary disease		FCZ 12 MKDay daily (max 800 mg)	FCZ alone Alternative: itraconazole

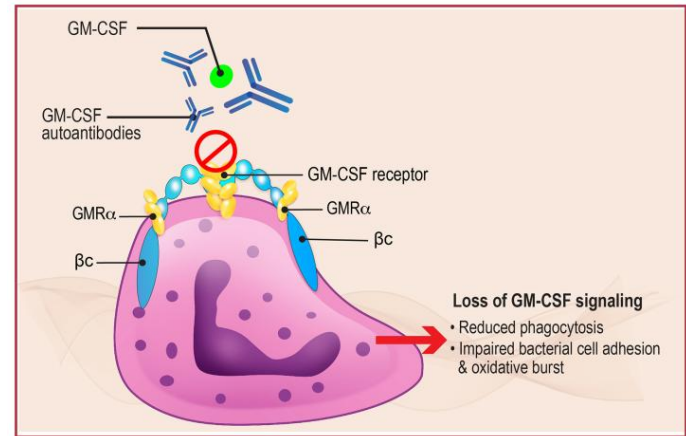
I; Induction, C; Consolidation, M; Maintenance, AMB; Amphotericin B, L-AMB; Liposomal amphotericin B, FCZ; Fluconazole

Work up for immune deficiency

- Autoantibodies to granulocyte macrophage colony-stimulating factor (anti-GM-CSF) → associated with *C. gatti*
- Idiopathic CD4 lymphopenia (ICL)
- X-linked hyper-IgM syndrome

Anti-GM-CSF autoantibodies

- GM-CSF normally binds to the GM-CSF receptor, present on the surface of **neutrophils and alveolar macrophages**, → initiate downstream signaling that regulates multiple functions
 - Phagocytosis, bacterial cell adhesion, and oxidative burst.
- High levels of AutoAbs that neutralize GM-CSF signaling cause **autoimmune pulmonary alveolar proteinosis (aPAP)**,
- accumulation of excess surfactant in the alveoli, leading to pulmonary insufficiency
- Serum anti-GM-CSF autoantibodies are reported as a predisposing factor for cryptococcosis.



Idiopathic CD4 lymphocytopenia (ICL)

“a documented absolute CD4 T lymphocyte count of **less than 300 cells per cubic millimeter** or of **less than 20% of total T cells** on more than one occasion, **no evidence of infection on HIV testing**, and the absence of any defined immunodeficiency or therapy associated with depressed levels of CD4 T cells.”

- **Infections at presentation**
 - Most common: Cryptococcal disease, persistent genital HPV infection, and nontuberculous mycobacterial infection
 - Others: herpes virus, hepatitis virus, tuberculosis, John Cunningham virus, and *P jirovecii* infection
- **Treatment:** Treating acute infection, preventing opportunistic infections
- **Prophylaxis Treatment Guideline:**
 - CD4 counts < 200 cells/mm³ → TMP/SMX
 - CD4 counts < 100 cells/mm³ → Azithromycin weekly
 - History of herpes infections → Acyclovir or valacyclovir
 - low levels of CD4⁺ and CD8⁺ cells → Anti-fungal treatment

Conclusion

- Cryptococcosis is a invasive fungal infection associated with significant morbidity and mortality
 - can be found in both immunocompetent and immunocompromised hosts
- Specific treatment and duration depends on populations and host status
 - Children, Adult, HIV, SOT, non-HIV, non-SOT
- Work up immune deficiency in patient with cryptococcosis
 - Anti-HIV, Anti-GM CSF, CD4 count
- Aware of antifungal resistance in patients with
- relapsed or persistent cryptococcosis.



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Thank you